MULTI_TOOLS PLATFORM SPECIFICATION

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1. Executive Overview

Multi_Tools delivers a high-performance, browser-first utility suite eliminating the need for remote cloud processi
Every operation including PDF manipulation, OCR text layer recognition, image optimization, and hashing
is executed entirely within client memory sandbox without external telemetry or data retention.

2. Component Architecture

The platform separates consumer tools from developer-focused utilities while adhering to standard security base

Module	Execution Model	Cache Policy	Status
PDF 31 Suite	Client-side pdf-lib WebAssembly	Ephemeral In-Memory	**Operational**
Optical OCR Engine	Browser Canvas & Glyph Mapping	Session Local	**Operational**
Image Studio	HTML5 Canvas 2D Context	24h Expiry TTL	**Operational**
Dev Regex & JSON	Local V8 Runtime Engine	Secure Session Only	**Operational**

3. Zero-Knowledge Privacy Standard

Security Exception Rules:

• Passwords and encryption keys are strictly omitted from localStorage persistence.

• Uploaded PDF documents, scanned images, and signatures stay within RAM.

• Temporary object URLs are revoked upon tool reset.

• 24-Hour TTL automatically invalidates non-sensitive calculation history.

4. System Verification & Sign-Off

This multi-page document confirms full test compatibility for Split PDF, Merge PDF, Organize PDF, Page Numbers, Watermarking, and Markdown extraction. All pages have been compiled according to ISO 32000-1 specifications.

Lead Systems Engineer:

Verified & Released for In-Browser Testing